

ServiceFlow SDM — Requirements Document

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1. Product overview

ServiceFlow SDM (Service Delivery Manager) is a full-stack platform for Cisco-powered enterprise service delivery operations targeting the hospitality vertical (MGM Resorts reference implementation). It unifies 17+ Cisco integration waves, Salesforce CRM, Digitized Delivery automation (Network as Code, Services as Code, Digital Document Solutions), and six SDC program consoles (PM, Service Delivery, Success, Renewals, Delivery Architect, Engineer) into a single operational surface.

1.1 Core value propositions

- **Single pane of glass** for multi-vendor, multi-domain service delivery across properties
- **Cisco API consolidation** — Waves 1–17 covering DNA Center, TAC, Smart Licensing, WebEx, ThousandEyes, Umbrella, Stealthwatch, DWDM, Cisco IQ, CSPC, ISE, Cisco Spaces, Secure Access, PSIRT/OpenVuln, Field Notices, FMC, and Salesforce CRM
- **Digitized Delivery** — NaC (9 solutions), SaC (12 waves A–J + AI Assistant + DDS), with ROI calculators and automation readiness scoring
- **SDC console parity** — replaces swivel-chair across vendor consoles with unified API-driven views
- **AI copilot integration** — context-rich prompt blocks for Microsoft 365 Copilot per view
- **Role-based experience** — each persona (PM, SDM, CXM, CDA, Engineer, HTOM) sees prioritized data

1.2 System components

Component	Technology	Purpose
Backend API	Express.js + TypeScript	REST API, auth, integrations, business logic
Frontend SPA	React 18 + MUI v6 + Vite	Authenticated operational dashboard
Mockup hub	Static HTML/CSS/JS	19-view interactive prototype with live data mode
Database	PostgreSQL 15	Persistent storage (schemas: <code>mgm</code> , <code>cisco</code> , <code>audit</code>)
Cache	Redis 7	Session/cache layer (connected at startup)
External APIs	Cisco, Salesforce, Microsoft, WebEx	Integration sources

2. Stakeholders and personas

2.1 Application roles (RBAC)

Role	Access level	Primary use cases
<code>admin</code>	Full access	System configuration, user management, all operations
<code>sdm</code>	Delivery operations	Incident management, device operations, source admin, sync
<code>tam</code>	Technical advisory	Incident triage, TAC linkage, source testing, sync
<code>csm</code>	Customer success	Read access, Salesforce case creation
<code>engineer</code>	Technical execution	Device management, incident updates, Salesforce read
<code>manager</code>	Program oversight	Properties, source admin, sync, Salesforce write
<code>viewer</code>	Read-only	Dashboard viewing, Salesforce read

2.2 SDC program consoles (mapped to views)

Console	Primary persona(s)	Key data sources
PM Console	PM, Manager	Salesforce Accounts + Opportunities, wave maturity, properties
Service Delivery Console	SDM, Engineer	Incidents, TAC SRs, Salesforce Cases, device health
Success Console	CXM, CSM	Salesforce Entitlements + Account health, adoption metrics
Renewals Console	CXM, Manager	Salesforce Opportunities + Service Contracts, pipeline
Delivery Architect Console	CDA	NaC/SaC architecture patterns, reference designs
Engineer Console	Engineer	Device operations, NaC pipelines, FMC/ISE config

3. System architecture

3.1 Runtime topology



3.2 Backend middleware stack (ordered)

1. **Helmet** (path-based CSP via `helmetByPath`)
2. **CORS** (from `CORS_ORIGIN` , credentials enabled)
3. **Cookie parser** (signed with `JWT_SECRET`)
4. **JSON body parser** (1 MB limit)
5. **Static file server** (`public/` directory)

3.3 Route mount map

Mount path	Router	Source
<code>/api/v1/health</code>	<code>healthRouter</code>	<code>routes/health.ts</code>
<code>/api/v1/auth</code>	<code>authRouter</code>	<code>routes/auth.ts</code> + <code>routes/sso.ts</code>
<code>/api/v1/properties</code>	<code>propertiesRouter</code>	<code>routes/properties.ts</code>
<code>/api/v1/devices</code>	<code>devicesRouter</code>	<code>routes/devices.ts</code>
<code>/api/v1/incidents</code>	<code>incidentsRouter</code>	<code>routes/incidents.ts</code>

Mount path	Router	Source
/api/v1/tac-cases	tacRouter	routes/tac.ts
/api/v1/integrations	integrationsRouter	routes/integrations.ts
/api/v1/admin	sourceAdminRouter	routes/sourceAdmin.ts
/api/v1/analytics/powerbi	powerBiRouter	routes/powerBi.ts
/api/v1/salesforce	salesforceRouter	routes/salesforce.ts

4. Authentication and authorization

4.1 Local authentication (JWT)

Requirement	Detail
REQ-AUTH-001	Users authenticate via <code>POST /api/v1/auth/login</code> with email + password
REQ-AUTH-002	Passwords validated with bcrypt; generic "Invalid username or password" on failure
REQ-AUTH-003	On success, API returns <code>accessToken</code> (JWT, default 24h) and <code>refreshToken</code> (default 7d)
REQ-AUTH-004	JWT signed with <code>JWT_SECRET</code> (minimum 32 characters); payload contains <code>userId</code> (UUID) and <code>role</code>
REQ-AUTH-005	Token refresh via <code>POST /api/v1/auth/refresh</code> with valid refresh token
REQ-AUTH-006	Logout via <code>POST /api/v1/auth/logout</code> (authenticated, returns 204)
REQ-AUTH-007	All protected routes require <code>Authorization: Bearer <token></code> header
REQ-AUTH-008	Role-based access enforced via <code>requireRoles([...])</code> middleware; returns 403 on mismatch

4.2 SSO / OIDC authentication (optional)

Requirement	Detail
REQ-SSO-001	OIDC SSO enabled via <code>SSO_ENABLED=true</code> with issuer discovery
REQ-SSO-002	Authorization Code flow with PKCE; state stored in signed httpOnly cookie
REQ-SSO-003	Callback matches user by email in <code>mgm.users</code> ; returns 403 if no matching account

Requirement	Detail
REQ-SSO-004	Optional JIT provisioning via <code>SSO_JIT_DEFAULT_ROLE</code> (creates user with bcrypt-hashed random password)
REQ-SSO-005	On success, redirects to <code>SSO_SUCCESS_REDIRECT</code> with tokens in URL hash fragment
REQ-SSO-006	Cross-field validation: when SSO is enabled, issuer, client ID, and redirect URI are required

4.3 Seed users (development)

Email	Role	Default password
<code>admin@serviceflow.local</code>	<code>admin</code>	<code>ChangeMe123!</code>
<code>sdm@serviceflow.local</code>	<code>sdm</code>	<code>ChangeMe123!</code>
<code>engineer@serviceflow.local</code>	<code>engineer</code>	<code>ChangeMe123!</code>

5. Database requirements

5.1 PostgreSQL configuration

Requirement	Detail
REQ-DB-001	PostgreSQL 15+ with extensions: <code>uuid-oss</code> , <code>pgcrypto</code> , <code>pg_trgm</code>
REQ-DB-002	Three schemas: <code>mgm</code> (application), <code>cisco</code> (vendor), <code>audit</code> (activity logging)
REQ-DB-003	Connection pool max 10 connections via <code>pg.Pool</code>
REQ-DB-004	Migration via <code>infra/migrations/001_init.sql</code> ; seed via <code>infra/seeds/001_seed.sql</code>

5.2 Core tables

Schema	Table	Purpose	Key columns
<code>mgm</code>	<code>users</code>	User accounts and RBAC	<code>user_id</code> (UUID), <code>email</code> (unique), <code>role</code> (enum), <code>password_hash</code>
<code>mgm</code>	<code>properties</code>	Managed properties/sites	<code>property_id</code> (UUID), <code>name</code> , <code>property_type</code> (hotel/casino/resort/venue)
<code>mgm</code>	<code>devices</code>	Network device inventory	<code>device_id</code> (UUID), <code>property_id</code> (FK), <code>hostname</code> , <code>ip_address</code> (INET), <code>serial_number</code> , <code>status</code> (enum), <code>dna_device_id</code> , <code>dna_managed</code> , <code>health_score</code>

Schema	Table	Purpose	Key columns
mgm	incidents	Incident lifecycle	incident_id (UUID), incident_number (sequence), priority (P1-P4), status (8-value enum), tac_case_number, tac_severity
mgm	incident_updates	Incident timeline entries	update_id, incident_id (FK), user_id (FK), content
cisco	tac_cases	TAC service request sync	tac_case_id (UUID), case_number (unique), severity (1-4), status, incident_id (FK)
cisco	technologies	Technology catalog	tech_id, name, category, vendor
cisco	licenses	License inventory	license_id, tech_id (FK), type, quantity, expiry
audit	activity_log	Audit trail	log_id, event_type, user_id, ip_address, user_agent, metadata (JSONB), created_at
mgm	integration_source_configs	Integration source registry (runtime-created)	source_name (PK), enabled, base_url, auth_type, schedule, credentials_ref, notes

5.3 Additional tables

mgm.property_technology_adoption, mgm.adoption_history, mgm.property_license_usage, mgm.sla_metrics, mgm.changes, mgm.teams_channels, mgm.webex_spaces, mgm.notifications, mgm.dashboards, mgm.reports, mgm.report_history, cisco.license_history, cisco.tac_case_updates

5.4 Sequences and triggers

- mgm.incident_number_seq — auto-generates INC-NNNNN on incident insert
- mgm.change_number_seq — auto-generates change numbers
- Resolution time trigger on incident status → resolved / closed
- updated_at auto-update triggers on major tables

6. Backend API requirements

6.1 Health check

ID	Method	Path	Auth	Description
REQ-API-001	GET	/api/v1/health	None	Returns { status: "ok" } after SELECT 1 on database pool

6.2 Properties

ID	Method	Path	Auth	Roles	Description
REQ-API-010	GET	/api/v1/properties	Bearer	Any	List all properties with property_id , name , property_type
REQ-API-011	POST	/api/v1/properties	Bearer	admin, sdm, manager	Create property; body: name (min 1), propertyType (enum); audit logged

6.3 Devices

ID	Method	Path	Auth	Roles	Description
REQ-API-020	GET	/api/v1/devices	Bearer	Any	List all devices with inventory fields
REQ-API-021	POST	/api/v1/devices	Bearer	admin, sdm, engineer, manager	Create device; body: propertyId (UUID), hostname , ipAddress (IP), serialNumber (min 3), status (enum); audit logged

6.4 Incidents

ID	Method	Path	Auth	Roles	Description
REQ-API-030	GET	/api/v1/incidents	Bearer	Any	List all incidents with full fields
REQ-API-031	POST	/api/v1/incidents	Bearer	admin, sdm, tam, csm, engineer, manager	Create incident; body: propertyId , optional deviceId , title (min 5), description (min 5), priority (P1-P4); auto-generates incident number; audit logged
REQ-API-032	PATCH	/api/v1/incidents/:id	Bearer	admin, sdm, tam, engineer, manager	Update incident status (8-value enum); audit logged
REQ-API-	POST	/api/v1/incidents/:id/updates	Bearer	Any authenticated	Add timeline update; body: content (min

ID	Method	Path	Auth	Roles	Description
033					1); audit logged
REQ-API-034	POST	/api/v1/incidents/:id/tac-link	Bearer	admin, sdm, tam, engineer	Link TAC case; body: tacCaseNumber (min 3), tacSeverity (1-4); audit logged

6.5 TAC cases

ID	Method	Path	Auth	Roles	Description
REQ-API-040	GET	/api/v1/tac-cases	Bearer	Any	List synced TAC cases from cisco.tac_cases

6.6 Integrations

ID	Method	Path	Auth	Roles	Description
REQ-API-050	POST	/api/v1/integrations/webex/war-room	Bearer	admin, sdm, tam, manager, engineer	Create WebEx space; body: optional title (max 200); returns roomId, title, webUrl
REQ-API-051	POST	/api/v1/integrations/sync/:source	Bearer	admin, sdm, tam, manager	Trigger sync for dna-center, tac, smart-licensing, or salesforce

6.7 Source administration

ID	Method	Path	Auth	Roles	Description
REQ-API-060	GET	/api/v1/admin/sources	Bearer	admin, sdm, manager	List all 18 integration source configurations
REQ-API-061	PUT	/api/v1/admin/sources/:name	Bearer	admin, sdm, manager	Update source config; body: enabled, baseUrl, authType, schedule, credentialsRef, notes (all optional)
REQ-API-062	POST	/api/v1/admin/sources/:name/test	Bearer	admin, sdm, tam, manager	Test source connection (runs actual sync/test for dna-center, tac, smart-licensing, salesforce)

6.8 Power BI analytics

ID	Method	Path	Auth	Roles	Description
REQ-API-070	GET	/api/v1/analytics/powerbi/embed	Bearer	All roles	Returns Power BI embed config when <code>POWERBI_ENABLED=true</code> ; otherwise <code>{ enabled: false }</code>

6.9 Salesforce CRM (see also §11)

Covered in detail in Section 11. Summary: 12 endpoints covering status, 8 object queries (cases, accounts, contacts, opportunities, entitlements, service-contracts, tasks, knowledge), console-summary aggregation, and case create/update.

7. External integration requirements

7.1 Integration wave registry

The system supports 18 named integration sources with the following wave assignments:

Wave	Source	Auth method	Sync schedule	API
1	DNA Center (Catalyst Center)	Basic → token	* /30 * * * *	REST: inventory, sites, health, issues
2	TAC Service Requests	API key + secret	* /15 * * * *	REST v2: cases, updates
3	Smart Licensing (SSM)	OAuth2 CC	0 * * * *	SWAPI: licenses, entitlements
4	WebEx Control Hub	Bot token	Manual	REST: rooms, spaces, meetings
4	Support API (Contract/EoX/Bug)	API key + secret	0 2 * * *	REST: SN2Contract, EoX, Bug
5	ThousandEyes	API token	* /15 * * * *	REST: tests, alerts, metrics
6	Umbrella	API key + OAuth	0 * * * *	REST: DNS/SIG policies, reports
7	Stealthwatch (SNA)	API basic/token	* /30 * * * *	REST: flow, security analytics
8	DWDM/Optical EMS	Collector/EMS	0 * * * *	Transport health
9	Cisco IQ	OAuth2/API key	0 4 * * *	REST: insights, entitlements
10	CSPC Collector	Collector registration	0 1 * * *	Collection, inventory, licensing
11	ISE + ISE as Code	ERS + pxGrid	* /30 * * * *	ERS/OpenAPI/pxGrid: identity, posture
12	Cisco Spaces	OAuth2/API key	0 * * * *	REST: location analytics
13	Secure Access	OAuth2 CC	* /15 * * * *	REST: SSE/ZTNA policies

Wave	Source	Auth method	Sync schedule	API
14	PSIRT/OpenVuln	OAuth2 CC	0 3 * * *	REST: advisories, CVEs
15	Field Notices	API key/OAuth	0 4 * * *	REST: FN feed, PID/serial/SW match
16	FMC (Firepower)	API token/basic	*/30 * * * *	REST: FTD inventory, policies
17	Salesforce CRM	OAuth2 password	*/15 * * * *	REST v59.0: Cases, Accounts, etc.

7.2 Implemented integration clients

Client	File	Status	Methods
DnaCenterClient	integrations/dnaCenterClient.ts	Active	listDevices() → upserts mgm.devices
TacClient	integrations/tacClient.ts	Active	listCases() → upserts cisco.tac_cases
SmartLicensingClient	integrations/smartLicensingClient.ts	Active	getEntitlements() → count only
WebexClient	integrations/webexClient.ts	Active	createRoom(title) → space creation
SupportApiClient	integrations/supportApiClient.ts	Stub	getContractBySerial, getEoxByPid, getBugById → null
SalesforceClient	integrations/salesforceClient.ts	Active	Full CRUD (see §11)

7.3 Sync jobs

Job	Trigger	Behavior
DNA Center sync	POST /sync/dna-center	Fetches devices, upserts into mgm.devices with dna_managed=true
TAC sync	POST /sync/tac	Fetches cases, upserts into cisco.tac_cases
Smart Licensing sync	POST /sync/smart-licensing	Fetches entitlements, returns count (no DB write)
Salesforce sync	POST /sync/salesforce	Tests connection, returns status

7.4 Meraki and AppDynamics (inline)

Requirement	Detail
REQ-INT-080	Meraki Dashboard data appears inline on Devices, Properties, and Integrations views (DD wave DD-K)

Requirement	Detail
REQ-INT-081	AppDynamics APM data appears inline on Devices, Incidents, and Overview views (DD wave DD-L)
REQ-INT-082	Neither Meraki nor AppDynamics has a standalone page; data is contextual within existing views

7.5 Power BI integration

Requirement	Detail
REQ-INT-090	Power BI embed via Microsoft Graph: Azure AD app registration with client credentials
REQ-INT-091	Backend generates embed token via <code>services/powerBiEmbed.ts</code> ; frontend renders via <code>powerbi-client</code> SDK
REQ-INT-092	Requires <code>POWERBI_ENABLED=true</code> and all 5 Power BI env vars (tenant, client, secret, workspace, report)
REQ-INT-093	Graceful fallback: when disabled, UI shows configuration instructions

8. Frontend application requirements

8.1 Technology stack

Requirement	Detail
REQ-FE-001	React 18 with TypeScript, Material UI v6, Vite build tool
REQ-FE-002	Dev server on port 3001 (<code>vite --host 0.0.0.0 --port 3001</code>)
REQ-FE-003	API base URL configurable via <code>VITE_API_BASE_URL</code> (defaults to <code>http://localhost:3000/api/v1</code>)
REQ-FE-004	JWT stored in <code>localStorage</code> as <code>accessToken</code>

8.2 Routes and pages

Route	Page component	Auth required	Description
<code>/login</code>	<code>LoginPage</code>	No	Email/password form; dev mode prefills seed credentials
<code>/dashboard</code>	<code>DashboardPage</code>	Yes	4 static KPI cards (Open Incidents, P1/P2, TAC Cases, Device Health)
<code>/incidents</code>	<code>IncidentsPage</code>	Yes	Fetches <code>GET /incidents</code> ; renders list with incident number, title, priority, status

Route	Page component	Auth required	Description
/devices	DevicesPage	Yes	Fetches GET /devices ; renders list with hostname, IP, status
/properties	PropertiesPage	Yes	Fetches GET /properties ; renders list with name, type
/powerbi-pm	PowerBiPmDashboardPage	Yes	Fetches Power BI embed config; renders embedded report or fallback
/salesforce	SalesforcePage	Yes	Fetches 6 Salesforce endpoints; KPI strip + 5 tabbed data tables
*	Redirect to /dashboard	—	Catch-all redirect

8.3 API client functions

Function	HTTP	Path	Purpose
login(email, password)	POST	/auth/login	Authenticate, store token
apiGet<T>(path)	GET	{path}	Generic authenticated GET
apiPost<T>(path, body)	POST	{path}	Generic authenticated POST
apiPatch<T>(path, body)	PATCH	{path}	Generic authenticated PATCH
fetchPowerBiEmbed()	GET	/analytics/powerbi/embed	Power BI embed config

9. Mockup hub requirements

9.1 Overview

Requirement	Detail
REQ-MOCK-001	Static HTML/CSS/JS mockup at backend/public/mockup/index.html
REQ-MOCK-002	19 view sections navigable via left sidebar with hash-based routing
REQ-MOCK-003	Three-column resizable layout: sidebar, main content, AI copilot pane
REQ-MOCK-004	Light/dark/system/daylight theme switching with geolocation-based auto-daylight
REQ-MOCK-005	Contrast modes: standard, soft, high
REQ-MOCK-006	Readability controls: text scale (A-/A/A+/A++), reduced motion toggle

Requirement	Detail
REQ-MOCK-007	Navigation tab reorder via drag-and-drop with localStorage persistence
REQ-MOCK-008	Main content block reorder via drag-and-drop per view
REQ-MOCK-009	AI copilot pane with per-view contextual insights, suggested questions, and Copilot clipboard export

9.2 View inventory (19 views)

#	View ID	Title	Key features
1	overview	Overview	KPIs (8 metrics), quick actions, timeline, sync cards, DD callout, SF pulse
2	sentiment	Sentiment & VoC	CSAT/NPS, signal blend, escalation queue, VoC API table
3	journeys	Journey signals	Active journeys, friction score, telemetry, analytics APIs
4	cx-command	Experience command	Financial KPIs, 5 next-step cards, customer financial table, trends, AI analysis, SF exec pulse
5	cx-role-actions	CX role actions	Role-based action matrix with DD/Meraki/AppDynamics columns
6	powerbi-pm	Power BI · Global PM	Embed capabilities, PM surfaces table, mock tiles, SF role matrix
7	incidents	Incidents	Queue metrics, incident table, detail preview, TAC correlation, DD/SF callouts
8	devices	Devices	Inventory table, assurance issues, Meraki table, AppDynamics table, DD/SF callouts
9	properties	Properties	Property cards with filters, site mapping, tech adoption, DD readiness, SF account health
10	integrations	Waves & integrations	Advisor engine, wave cards, API builds table, DD wave registry
11	mvp-journey	MVP journey & adoption	Journey map, CSPC/IQ deep panels, DD by phase, SF pipeline by phase
12	sources	Source administration	Source cards, change log, env checklist, DD source readiness, SF source detail
13	consoles	Console ↔ wave map	Console-wave matrix, inline integrations table, DD parity, SF CRM layer
14	sdcroles	SDC personas & consoles	6 persona cards, next-steps engine, 6 console cards, renewal table, persona matrix, DD recommendations, SF feeds
15	security	Security (PSIRT)	OpenVuln metrics, advisories table, remediation timeline, DD callout

#	View ID	Title	Key features
16	fieldnotes	Field notices	FN metrics, FN table, bulk actions, DD remediation, SF case tracking
17	network-as-code	Network as Code	9 NaC solutions, API matrix, adoption framework, ROI calculator (\$886K)
18	services-as-code	Services as Code	12 SaC waves, partner table, Cisco Live resources, ROI calculator (\$1,258K)
19	digital-document-solutions	Digital Document Solutions	Document lifecycle, 12 templates, DDS API table

9.3 Live mode

Requirement	Detail
REQ-MOCK-020	Toggle live data mode via <code>?live=1</code> or UI toggle
REQ-MOCK-021	Requires valid JWT in localStorage (same origin or pasted token)
REQ-MOCK-022	Fetches <code>GET /properties</code> , <code>/devices</code> , <code>/incidents</code> , <code>/tac-cases</code> , <code>/admin/sources</code>
REQ-MOCK-023	Overwrites mock KPIs, tables, and property cards with live API data
REQ-MOCK-024	Supports war-room creation via <code>POST /integrations/webex/war-room</code>

9.4 Integration advisor

Requirement	Detail
REQ-MOCK-030	AI-driven integration advisor on the Waves & Integrations view
REQ-MOCK-031	Merges <code>APP_DATA_SNAPSHOT</code> with live DOM KPIs when in live mode
REQ-MOCK-032	Generates up to 6 prioritized action steps and 8 wave card recommendations
REQ-MOCK-033	Rules: P1/P2 incidents, observability gaps, health <80, stale telemetry, PSIRT/FN exposure, journey friction, ISE status

9.5 AI copilot pane

Requirement	Detail
REQ-MOCK-040	Each of 19 views has <code>copilotContextPlain</code> , <code>blocks[]</code> , and <code>suggestions[]</code> in <code>AI_INSIGHTS_BY_VIEW</code>
REQ-MOCK-041	Blocks support <code>urgent: true</code> (magenta styling) and <code>dd: true</code> (Digitized Delivery styling)
REQ-MOCK-042	"Copy for Copilot" builds context block with view title, descriptor, insights, and user question

Requirement	Detail
REQ-MOCK-043	"Open Copilot" links to <code>microsoft365.com/chat</code> for Microsoft 365 Copilot
REQ-MOCK-044	Suggested questions auto-populate the query field on click

9.6 Theme and accessibility

Requirement	Detail
REQ-MOCK-050	Theme: dark (default), light, system-match, daylight-match (via sunrise-sunset.org API + geolocation)
REQ-MOCK-051	Daylight auto-refreshes every 30 minutes based on GPS coordinates
REQ-MOCK-052	Contrast: standard, soft (reduced border/shadow), high (increased contrast)
REQ-MOCK-053	Text scale: 4 levels via <code>--text-scale</code> CSS variable
REQ-MOCK-054	Reduced motion: respects <code>prefers-reduced-motion</code> and manual toggle
REQ-MOCK-055	Navigation density slider for sidebar spacing
REQ-MOCK-056	Brand logo automatically swaps light/dark variant based on theme

9.7 Data structures

Object	Count	Purpose
MGM_PROPERTIES	18 properties	Mock portfolio (11 Las Vegas, 7 remote)
INTEGRATION_SOURCES	18 sources	Source registry mock data
CONSOLE_MAP_ROWS	21 rows	Console-to-wave mapping matrix
MVP_JOURNEY_BY_WAVE	16 waves (W1-W16)	Journey map with phase, maturity, and recommendations
APP_DATA_SNAPSHOT	1 object	Unified mock KPIs for personas and advisor

10. Digitized Delivery requirements

10.1 Network as Code (NaC)

Requirement	Detail
REQ-DD-001	9 NaC solutions tracked: Catalyst Center, SD-WAN, ISE, IOS-XE, and 5 NaC tools

Requirement	Detail
REQ-DD-002	42% automation readiness metric across 1,626 devices (DNA + Meraki)
REQ-DD-003	ROI calculator with 8 configurable inputs; estimated 3-year ROI \$886K
REQ-DD-004	4-phase adoption framework: Assess → Pilot → Scale → Operate
REQ-DD-005	NaC API matrix showing Terraform providers, Ansible collections, pyATS tests per solution

10.2 Services as Code (SaC)

Requirement	Detail
REQ-DD-010	12 SaC waves (A–J + AI Assistant + DDS)
REQ-DD-011	Wave cards: ACI, SD-WAN, Wireless, ISE, Meraki, Catalyst Center, ThousandEyes, FMC, IOS-XE, Webex, AI Assistant, DDS
REQ-DD-012	14 technologies, 7 dev waves, 38% coverage, 3 partner routes
REQ-DD-013	ROI calculator; estimated 3-year ROI \$1,258K
REQ-DD-014	Partner integration table and Cisco Live resources

10.3 Digital Document Solutions (DDS)

Requirement	Detail
REQ-DD-020	12 document templates across 4 lifecycle phases
REQ-DD-021	Template categories: assessment, runbook, incident summary, QBR pack
REQ-DD-022	DDS API summary table

10.4 DD integration across views

Requirement	Detail
REQ-DD-030	Every DD callout uses CSS class <code>dd-callout</code> with magenta (Splunk) accent color
REQ-DD-031	DD callouts present on 12 of 19 views: overview, cx-role-actions, incidents, devices, properties, integrations, mvp-journey, sources, consoles, sdcroles, security, fieldnotes
REQ-DD-032	DD automation waves DD-A through DD-N mapped in the integration wave registry
REQ-DD-033	AI insight blocks with <code>dd: true</code> flag present in 11 views
REQ-DD-034	Meraki as Code (DD-K) and AppDynamics (DD-L) tracked as DD waves with metrics

11. Salesforce CRM integration requirements

11.1 Authentication

Requirement	Detail
REQ-SF-001	OAuth 2.0 password grant via Salesforce Connected App
REQ-SF-002	Token cached with 110-minute TTL; auto-invalidation on 401
REQ-SF-003	Requires: SALESFORCE_ENABLED , CLIENT_ID , CLIENT_SECRET , USERNAME , PASSWORD
REQ-SF-004	Optional SECURITY_TOKEN appended to password

11.2 API endpoints

ID	Method	Path	Roles	Description
REQ-SF-010	GET	/salesforce/status	Any authenticated	Test connection; returns ok , instanceUrl , orgId
REQ-SF-011	GET	/salesforce/cases	All SF roles	Query Cases with limit (max 500)
REQ-SF-012	GET	/salesforce/accounts	All SF roles	Query Accounts
REQ-SF-013	GET	/salesforce/contacts	All SF roles	Query Contacts; optional accountId filter (validated: 15-18 alphanumeric)
REQ-SF-014	GET	/salesforce/opportunities	All SF roles	Query Opportunities
REQ-SF-015	GET	/salesforce/entitlements	All SF roles	Query Entitlements
REQ-SF-016	GET	/salesforce/service-contracts	All SF roles	Query Service Contracts
REQ-SF-017	GET	/salesforce/tasks	All SF roles	Query open Tasks
REQ-SF-018	GET	/salesforce/knowledge	All SF roles	Query published Knowledge articles (max 200)

ID	Method	Path	Roles	Description
REQ-SF-019	GET	/salesforce/console-summary	All SF roles	Aggregated metrics for PM, Delivery, Success, Renewals, Architect consoles
REQ-SF-020	POST	/salesforce/cases	Write roles	Create Case with validated fields: <code>subject</code> (required, max 500), <code>description</code> (max 5000), <code>priority</code> (enum), <code>accountId</code> / <code>contactId</code> (SF ID format)
REQ-SF-021	PATCH	/salesforce/cases/:caseId	Write roles	Update Case; allowed fields: Status, Priority, Description, Subject; <code>caseId</code> validated as SF ID format

11.3 Salesforce objects queried

Object	SOQL fields	Console mapping
Case	Id, CaseNumber, Subject, Status, Priority, Type, Origin, Account.Name, Contact.Name, Owner.Name, CreatedDate, ClosedDate	Service Delivery, PM
Account	Id, Name, Type, Industry, Phone, Website, BillingCity, BillingState, Owner.Name, AnnualRevenue	PM, Success, HTOM
Contact	Id, FirstName, LastName, Email, Phone, Title, Department, Account.Name	PM, Success
Opportunity	Id, Name, StageName, Amount, CloseDate, Probability, Type, AccountId, Owner.Name, NextStep, FiscalYear	Renewals, PM
Entitlement	Id, Name, AccountId, Account.Name, StartDate, EndDate, Status, Type, RemainingCases	Success, Renewals
ServiceContract	Id, Name, AccountId, Account.Name, StartDate, EndDate, Status, Term, ApprovalStatus	Renewals
Task	Id, Subject, Status, Priority, ActivityDate, Owner.Name, Type	All consoles
Knowledge__kav	Id, Title, ArticleNumber, PublishStatus, VersionNumber	Knowledge base

11.4 Console summary aggregation

The `/console-summary` endpoint computes:

Console	Metrics
PM	<code>totalAccounts</code> , <code>activeOpportunities</code> , <code>pipelineValue</code> , <code>closedWonCount</code> , <code>openTasks</code>
Delivery	<code>openCases</code> , <code>highPriorityCases</code> , <code>totalCases</code>
Success	<code>totalAccounts</code> , <code>activeEntitlements</code> , <code>openTasks</code>

Console	Metrics
Renewals	<code>totalEntitlements</code> , <code>activeEntitlements</code> , <code>expiringIn90Days</code> , <code>pipelineValue</code> , <code>pipelineCount</code>
Architect	<code>totalAccounts</code> , <code>totalCases</code>

11.5 SF panels in mockup hub

Salesforce CRM panels (`sf-panel`, blue accent `#00a1e0`) are present on 11 views: overview, cx-command, powerbi-pm, incidents, devices, properties, mvp-journey, sources, consoles, sdcroles, fieldnotes.

12. Infrastructure and deployment

12.1 Docker Compose

Service	Image	Port	Config
<code>postgres</code>	<code>postgres:15-alpine</code>	5432	DB: <code>serviceflow_sdm</code> , user: <code>serviceflow_admin</code> , volume: <code>pg_data</code>
<code>redis</code>	<code>redis:7-alpine</code>	6379	Default config
<code>backend</code>	Build from <code>./backend</code>	3000	<code>env_file</code> : <code>.env.example</code> , depends on postgres + redis
<code>frontend</code>	Build from <code>./frontend</code>	3001	<code>VITE_API_BASE_URL</code> = <code>http://localhost:3000/api/v1</code> , depends on backend

12.2 CI/CD workflows

Workflow	Trigger	Jobs
<code>ci.yml</code>	Push, PR	Backend: install, build, lint, test; Frontend: same
<code>deploy-mockup-pages.yml</code>	Push to main, manual	Deploy <code>backend/public/mockup/</code> to GitHub Pages
<code>scheduled-autofix.yml</code>	Monday 09:00 UTC, manual	<code>lint:fix</code> in backend + frontend; auto-PR if changes

12.3 Migration and seeding

```
npm run migrate    # Runs 001_init.sql + 001_seed.sql
npm run seed-passwords # Hashes dev passwords for seed users
```

13. Security requirements

13.1 Authentication security

Requirement	Detail
REQ-SEC-001	JWT secrets minimum 32 characters; validated at startup via Zod
REQ-SEC-002	Passwords hashed with bcrypt (configurable rounds, default 12)
REQ-SEC-003	Generic error messages on authentication failure (no account enumeration)
REQ-SEC-004	SSO state stored in signed httpOnly cookie with PKCE
REQ-SEC-005	Login page default credentials gated behind <code>import.meta.env.DEV</code>

13.2 Input validation

Requirement	Detail
REQ-SEC-010	Zod schema validation on all request bodies
REQ-SEC-011	Parameterized SQL queries throughout (no string concatenation)
REQ-SEC-012	Salesforce ID format validation (<code>/^[a-zA-Z0-9]{15,18}\$/</code>) on all SF ID inputs
REQ-SEC-013	String length bounds on user-supplied content (subject: 500, description: 5000, title: 200)
REQ-SEC-014	Priority values validated against allow-list enum

13.3 Transport and headers

Requirement	Detail
REQ-SEC-020	Helmet security headers on all responses (path-specific CSP)
REQ-SEC-021	CORS restricted to configured origins
REQ-SEC-022	JSON body size limited to 1 MB
REQ-SEC-023	Mockup CSP relaxed (allows <code>https:</code> and <code>wss:</code> for external resources)
REQ-SEC-024	Power BI CSP allows Microsoft embed domains

13.4 Error handling

Requirement	Detail
REQ-SEC-030	All async route handlers wrapped in try/catch with appropriate error codes
REQ-SEC-031	External API failures return 502; internal failures return 500
REQ-SEC-032	Error messages truncated (max 300 chars from upstream APIs)

14. Observability and audit

14.1 Audit logging

Requirement	Detail
REQ-AUDIT-001	Fire-and-forget <code>INSERT</code> into <code>audit.activity_log</code> for security-sensitive operations
REQ-AUDIT-002	Logged fields: <code>event_type</code> , <code>user_id</code> , <code>ip_address</code> , <code>user_agent</code> , <code>metadata</code> (JSONB with method, path)
REQ-AUDIT-003	Audit events: <code>property.create</code> , <code>device.create</code> , <code>incident.create</code> , <code>incident.status.update</code> , <code>incident.update.create</code> , <code>incident.tac.link</code>

14.2 Health monitoring

Requirement	Detail
REQ-AUDIT-010	Health endpoint verifies database connectivity (<code>SELECT 1</code>)
REQ-AUDIT-011	Redis connects at startup with non-fatal failure (<code>.catch(() => undefined)</code>)

15. Non-functional requirements

15.1 Performance

Requirement	Detail
REQ-NFR-001	Database connection pool: max 10 connections
REQ-NFR-002	Salesforce token cache with 110-minute TTL to minimize auth round-trips
REQ-NFR-003	Salesforce query limits enforced server-side (max 500 records per query)
REQ-NFR-004	Console-summary endpoint parallelizes 5 Salesforce queries via <code>Promise.all</code>

15.2 Accessibility

Requirement	Detail
REQ-NFR-010	Mockup hub supports screen reader announcements via <code>aria-live</code> regions
REQ-NFR-011	Keyboard navigation for sidebar (<code>Alt+↑/↓</code>), pane resize (arrows), and view focus
REQ-NFR-012	Skip-to-content link
REQ-NFR-013	<code>prefers-reduced-motion</code> respected and manually toggleable
REQ-NFR-014	4-level text scaling for readability

15.3 Browser support

Requirement	Detail
REQ-NFR-020	Mockup hub uses CSS custom properties, grid, flexbox (modern browsers)
REQ-NFR-021	Power BI embed requires powerbi-client SDK (modern browsers)

15.4 Development experience

Requirement	Detail
REQ-NFR-030	Hot reload via <code>tsx watch</code> (backend) and <code>vite</code> (frontend)
REQ-NFR-031	TypeScript strict mode enabled
REQ-NFR-032	ESLint with TypeScript parser
REQ-NFR-033	Vitest for testing (framework configured, no tests written yet)

16. Out of scope (MVP)

- Advanced reporting beyond basic JSON/CSV export
- Mobile-native applications (iOS/Android)
- Multi-region / high-availability database topology
- Standalone Meraki Dashboard or AppDynamics pages (inline only)
- Real-time WebSocket push notifications
- Email notification system
- Microsoft Teams integration (env vars defined, not implemented)
- ThousandEyes, Umbrella, Stealthwatch, DWDM, Cisco IQ, CSPC, Cisco Spaces, Secure Access direct API wiring (source config registry and mockup data exist; no live sync jobs)

17. Milestones and definition of done

17.1 Functional milestones

#	Milestone	Description
1	Bootstrap	Project scaffold, Docker Compose, CI pipeline
2	Database + seed	Schema migration, seed data, dev password hashing
3	Core APIs	Properties, devices, incidents (CRUD + timeline), TAC linkage
4	Frontend	React SPA with login, dashboard, list pages
5	DNA + TAC	DNA Center device sync, TAC case sync, live data in UI
6	Waves 5–16	Source registry, connection testing, mockup data for all waves

#	Milestone	Description
7	Meraki + AppDynamics	Inline data on Devices, Properties, Incidents views
8	Digitized Delivery	NaC/SaC/DDS surfaces, DD waves in registry, ROI calculators
9	Salesforce CRM (Wave 17)	Full CRUD, console-summary, all SDC console feeds
10	QA + hardening	Security review, audit, error handling, input validation
11	Deployment runbook	Non-prod deployment documentation, smoke tests

17.2 Definition of done

- ☐ `docker compose up` starts all services; health check returns OK
- ☐ Login with seed credentials; RBAC enforced on all protected routes
- ☐ Incidents: create, update status, add timeline, link TAC case
- ☐ Device inventory synced from at least one integration (DNA Center)
- ☐ CI passes: build, lint, type-check for both backend and frontend
- ☐ Per enabled integration: console-equivalent core tasks available via API
- ☐ Meraki/AppDynamics visible inline where specified (DD-K, DD-L)
- ☐ NaC/SaC/DDS surfaces with magenta accent, DD waves DD-A–DD-N in registry
- ☐ Salesforce powering all 6 SDC consoles, Wave 17 in source registry
- ☐ Experience Command console-summary with live SF aggregation
- ☐ 19-view mockup hub with DD callouts, SF panels, AI insights, and live mode
- ☐ Audit logging on security-sensitive operations
- ☐ All route handlers have try/catch error handling
- ☐ No hardcoded credentials in production builds

Appendix A — Environment variables

Full inventory of 50+ environment variables documented in `.env.example`:

App: `NODE_ENV`, `PORT`, `FRONTEND_PORT`, `APP_PUBLIC_URL`

Database: `DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASSWORD`

Redis: `REDIS_HOST`, `REDIS_PORT`

Security: `JWT_SECRET`, `JWT_REFRESH_SECRET`, `JWT_EXPIRE`, `JWT_REFRESH_EXPIRE`, `BCRYPT_ROUNDS`

CORS: `CORS_ORIGIN`, `SOCKET_IO_PATH`, `SOCKET_IO_CORS`

SSO/OIDC: `SSO_ENABLED`, `SSO_ISSUER`, `SSO_CLIENT_ID`, `SSO_CLIENT_SECRET`, `SSO_REDIRECT_URI`, `SSO_SUCCESS_REDIRECT`, `SSO_SCOPES`, `SSO_JIT_DEFAULT_ROLE`

TAC: `TAC_API_KEY`, `TAC_API_SECRET`, `TAC_BASE_URL`, `TAC_CONTRACT_NUMBER`

DNA Center: `DNA_CENTER_HOST`, `DNA_CENTER_USERNAME`, `DNA_CENTER_PASSWORD`, `DNA_CENTER_PORT`

Smart Licensing: SMART_LICENSEING_CLIENT_ID , SMART_LICENSEING_CLIENT_SECRET ,
SMART_LICENSEING_TOKEN_URL , SMART_LICENSEING_API_URL

WebEx: WEBEX_CLIENT_ID , WEBEX_CLIENT_SECRET , WEBEX_BOT_TOKEN , WEBEX_WEBHOOK_SECRET ,
WEBEX_REDIRECT_URI

Support APIs: CISCO_SUPPORT_API_KEY , CISCO_SUPPORT_API_SECRET

PSIRT: OPENVULN_CLIENT_ID , OPENVULN_CLIENT_SECRET , OPENVULN_TOKEN_URL

Field Notices: FIELD_NOTICE_API_BASE_URL , CISCO_FN_API_KEY

FMC: FMC_BASE_URL , FMC_API_TOKEN

Power BI: POWERBI_ENABLED , POWERBI_TENANT_ID , POWERBI_CLIENT_ID , POWERBI_CLIENT_SECRET ,
POWERBI_WORKSPACE_ID , POWERBI_REPORT_ID

Salesforce: SALESFORCE_ENABLED , SALESFORCE_LOGIN_URL , SALESFORCE_CLIENT_ID ,
SALESFORCE_CLIENT_SECRET , SALESFORCE_USERNAME , SALESFORCE_PASSWORD ,
SALESFORCE_SECURITY_TOKEN , SALESFORCE_API_VERSION

Feature flags: ENABLE_TAC_AUTO_CREATE , ENABLE_TEAMS_INTEGRATION , ENABLE_WEBEX_INTEGRATION ,
ENABLE_EMAIL_NOTIFICATIONS

Appendix B — Database schema

Schemas and extensions

```
CREATE EXTENSION IF NOT EXISTS "uuid-oss";  
CREATE EXTENSION IF NOT EXISTS "pgcrypto";  
CREATE EXTENSION IF NOT EXISTS "pg_trgm";  
CREATE SCHEMA IF NOT EXISTS mgm;  
CREATE SCHEMA IF NOT EXISTS cisco;  
CREATE SCHEMA IF NOT EXISTS audit;
```

Core table definitions

mgm.users — user_id UUID PK, email UNIQUE, full_name , role
(admin|sdm|tam|csm|engineer|manager|viewer), password_hash , timestamps

mgm.properties — property_id UUID PK, name , property_type (hotel|casino|resort|venue),
it_manager_user_id FK, timestamps

mgm.devices — device_id UUID PK, property_id FK, hostname , ip_address INET, ipv6_address ,
mac_address , serial_number , status (active|inactive|maintenance|decommissioned|failed),
dna_device_id , dna_managed BOOLEAN, health_score , contract/warranty/EOL fields, timestamps

mgm.incidents — incident_id UUID PK, incident_number UNIQUE (auto-sequence),
parent_incident_id self-FK, property_id FK, device_id FK, title , description , priority (P1-
P4), status (open|acknowledged|investigating|in-progress|pending|resolved|closed|cancelled),

reported_by FK, assigned_to FK, tac_case_number , tac_case_id FK, tac_severity (1-4), resolution_time , timestamps

mgm.incident_updates — update_id UUID PK, incident_id FK, user_id FK, content , created_at

cisco.tac_cases — tac_case_id UUID PK, case_number UNIQUE, severity (1-4), status , incident_id FK, last_update_at , timestamps

audit.activity_log — log_id UUID PK, event_type , user_id , ip_address , user_agent , metadata JSONB, created_at

mgm.integration_source_configs (runtime) — source_name TEXT PK, enabled BOOLEAN, base_url , auth_type , schedule , credentials_ref , notes , updated_by UUID, timestamps

Appendix C — API route inventory

Authentication (public)

Method	Path	Body/Query	Response
POST	/api/v1/auth/login	{ email, password }	{ accessToken, refreshToken }
POST	/api/v1/auth/refresh	{ refreshToken }	{ accessToken }
POST	/api/v1/auth/logout	—	204
GET	/api/v1/auth/sso/login	—	302 to IdP
GET	/api/v1/auth/sso/callback	OIDC params	302 to success URL

Core resources (Bearer JWT required)

Method	Path	Roles	Body/Query	Response
GET	/api/v1/health	—	—	{ status }
GET	/api/v1/properties	Any	—	Property[]
POST	/api/v1/properties	admin, sdm, manager	{ name, propertyType }	Property
GET	/api/v1/devices	Any	—	Device[]
POST	/api/v1/devices	admin, sdm, engineer, manager	{ propertyId, hostname, ipAddress, serialNumber, status }	Device
GET	/api/v1/incidents	Any	—	Incident[]

Method	Path	Roles	Body/Query	Response
POST	/api/v1/incidents	admin, sdm, tam, csm, engineer, manager	{ propertyId, deviceId?, title, description, priority }	Incident
PATCH	/api/v1/incidents/:id	admin, sdm, tam, engineer, manager	{ status }	{ incident_id, status }
POST	/api/v1/incidents/:id/updates	Any	{ content }	Update
POST	/api/v1/incidents/:id/tac-link	admin, sdm, tam, engineer	{ tacCaseNumber, tacSeverity }	{ incident_id, tac_case_number, tac_severity }
GET	/api/v1/tac-cases	Any	—	TacCase[]

Integrations and admin

Method	Path	Roles	Body/Query	Response
POST	/api/v1/integrations/webex/war-room	admin, sdm, tam, manager, engineer	{ title? }	{ roomId, title, webUrl }
POST	/api/v1/integrations/sync/:source	admin, sdm, tam, manager	—	{ source, processed, status? }
GET	/api/v1/admin/sources	admin, sdm, manager	—	SourceConfig[]
PUT	/api/v1/admin/sources/:name	admin, sdm, manager	Partial config	SourceConfig
POST	/api/v1/admin/sources/:name/test	admin, sdm, tam, manager	—	{ sourceName, status }
GET	/api/v1/analytics/powerbi/embed	All roles	—	PowerBiEmbed

Salesforce CRM

Method	Path	Roles	Body/Query	Response
GET	/api/v1/salesforce/status	Any	—	{ ok, instanceUrl?, orgId? }
GET	/api/v1/salesforce/cases	SF_ROLES	?limit=N	{ configured, totalSize,

Method	Path	Roles	Body/Query	Response
				<code>records }</code>
GET	<code>/api/v1/salesforce/accounts</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/contacts</code>	SF_ROLES	<code>?limit=N&accountId=X</code>	Same pattern
GET	<code>/api/v1/salesforce/opportunities</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/entitlements</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/service-contracts</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/tasks</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/knowledge</code>	SF_ROLES	<code>?limit=N</code>	Same pattern
GET	<code>/api/v1/salesforce/console-summary</code>	SF_ROLES	—	Aggregated console metrics
POST	<code>/api/v1/salesforce/cases</code>	SF_WRITE	<code>{ subject, description?, priority?, accountId?, contactId? }</code>	<code>{ id, success }</code>
PATCH	<code>/api/v1/salesforce/cases/:caseId</code>	SF_WRITE	<code>{ Status?, Priority?, Description?, Subject? }</code>	<code>{ id, updated }</code>